

Vellore Institute of Technology, Chennai Campus
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BEVD204L Electronic Circuits
Course Project Report
BJT Based Thermal Biasing System

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1 Objective

- To understand and analyze various BJT based thermal monitoring circuits for on chip thermal sensing.
- To adapt a suitable Thermal sensing circuit with temperature sensitive circuits such as a Class A Series Amplifier.

2 Introduction

Temperature sensing has become an essential component in modern integrated circuits, particularly in applications such as microprocessors, power management systems, automotive electronics, and biomedical devices. As device densities and power consumption increase, accurate on-chip thermal monitoring is necessary to ensure system reliability and performance. Among various temperature sensing techniques,[1] BJT-based temperature sensors are widely preferred due to their high accuracy, repeatability, and compatibility with standard CMOS fabrication processes. The operation of BJT-based temperature sensors is fundamentally based on the temperature sensitivity of the base-emitter junction. The base-emitter voltage decreases with increasing temperature, making it a reliable temperature-dependent parameter. In a Class A configuration, the transistor is biased to operate continuously in the active region, ensuring constant conduction and minimizing distortion. This mode of operation improves the linearity of the output signal and enhances stability under varying operating conditions, which is essential for accurate temperature measurement [2] [3].

3 Review of existing literature

This literature review focuses on the development and analysis of BJT-based temperature sensing circuits, particularly those utilizing PTAT (Proportional-To-Absolute-Temperature) techniques. It examines prior work on the temperature dependence of BJT characteristics, methods for generating PTAT signals using differential transistor configurations, and the application of bandgap principles in temperature-related circuits. The review also discusses the role of signal conditioning and amplification in improving output usability, along with the impact of non-idealities such as device mismatch and process variations. Additionally, recent advancements in low-power and high-accuracy temperature sensors are considered to highlight current trends and limitations, providing a foundation for understanding the design and implementation of the proposed temperature sensing system.

Existing literature on BJT-based temperature sensors demonstrates high accuracy and advanced performance; however, several important gaps remain. Many designs achieve excellent precision

through calibration and complex circuit techniques, as shown by Michiel A. P. Pertijs et al. [4] and Souri et al. [5], but these approaches increase design complexity, area, and cost, making them less suitable for simple or educational implementations. Foundational analyses by Yannis P. Tsividis [6] and practical works such as Wang et al. [7] highlight non-idealities in BJT behavior; however, there is limited emphasis on detailed investigation of PTAT slope variation and its impact on linearity across a wide temperature range. Furthermore, high-performance sensors like those proposed by Lai et al. [8] rely heavily on calibration and trimming, leaving a gap for calibration-free or minimally calibrated analog designs. The influence of analog signal conditioning stages is also underexplored, as classical texts such as Allen and Holberg [9] and Gray and Meyer [6] discuss amplifier design in general but do not explicitly analyze its effect on temperature-dependent outputs in sensing systems. Additionally, prior work by Robert J. Widlar [10] and Paul Brokaw [11][12] laid the theoretical foundation for temperature-dependent circuits, yet modern studies focus more on optimized end results rather than intermediate behavior such as PTAT generation and scaling. There is also a noticeable trade-off in literature between ultra-low-power designs, such as Aminzadeh et al. [13], and high-accuracy systems, such as Pertijs et al. [14], with limited exploration of balanced designs that achieve moderate accuracy with low complexity. Moreover, amplified or high-voltage temperature-dependent outputs are rarely discussed, as most works concentrate on low-voltage bandgap references. Therefore, there is a clear need for a simplified, simulation-driven BJT temperature sensing approach that emphasizes PTAT generation, slope analysis, analog amplification effects, and practical implementation without heavy reliance on calibration, thereby bridging the gap between theoretical concepts and real-world circuit behavior.

3.1 Research Gap

Existing literature on BJT-based temperature sensors demonstrates high accuracy and advanced performance; however, several important gaps remain. Many designs achieve excellent precision through calibration and complex circuit techniques, as shown by Michiel A. P. Pertijs et al. [14] and Souri et al. [9], but these approaches increase design complexity, area, and cost, making them less suitable for simple or educational implementations. Foundational analyses by Yannis P. Tsividis [15] and practical works such as Wang et al. [4] highlight non-idealities in BJT behavior; however, there is limited emphasis on detailed investigation of PTAT slope variation and its impact on linearity across a wide temperature range. Furthermore, high-performance sensors like those proposed by Lai et al. [7] rely heavily on calibration and trimming, leaving a gap for calibration-free or minimally calibrated analog designs.

4 Methodology/ Implementation

The proposed temperature sensing circuit is designed using a combination of Bipolar Junction Transistors (BJTs) and MOSFET-based current mirrors to generate a stable Proportional-To-Absolute-Temperature (PTAT) output. The system exploits the temperature-dependent base-emitter voltage of BJTs while ensuring stable biasing through current mirror configurations [16]. Additionally, Class A amplifier biasing is employed to maintain continuous operation in the active region, thereby improving linearity and signal stability. The overall methodology consists of: Temperature sensing using BJT

4.1 System Overview

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The overall methodology consists of temperature sensing using a BJT:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}} \quad (1)$$

$$V_T = \frac{kT}{q} \quad (2)$$

$$V_{BE} = V_T \ln \left(\frac{I_C}{I_S} \right) \quad (3)$$

The base-emitter voltage (V_{BE}) of a Bipolar Junction Transistor exhibits a well-defined dependence on temperature, characterized by a negative temperature coefficient:

$$\alpha = \frac{dV_{BE}}{dT} \approx -2 \text{ mV}/^\circ\text{C} \quad (4)$$

This indicates that V_{BE} decreases linearly with an increase in temperature under constant biasing conditions.

4.2 PTAT Voltage Generation

In PTAT (Proportional-To-Absolute-Temperature) voltage generation, two BJTs are operated at different current densities to produce a voltage difference between their base-emitter terminals:

$$\Delta V_{BE} = V_T \ln(n) \quad (5)$$

where the current ratio n is defined as:

$$n = \frac{I_{C1}}{I_{C2}} \quad (6)$$

The thermal voltage is given by:

$$V_T = \frac{kT}{q} \quad (7)$$

Substituting V_T into the PTAT equation:

$$\Delta V_{BE} = \frac{kT}{q} \ln(n) \quad (8)$$

Since $\frac{kT}{q}$ is directly proportional to absolute temperature T , it follows that:

$$\Delta V_{BE} \propto T \quad (9)$$

Thus, the generated voltage difference exhibits a linear dependence on temperature, resulting in a PTAT characteristic suitable for temperature sensing applications [18].

4.3 Application and Results

To further enhance the applications of the BJT thermal sensing circuit (Circuit 2), a Class A power amplifier is considered as a demonstration platform. The results show that, after adaptive biasing, there is a significant stabilization in voltage variation across different temperature ranges. This improvement highlights the potential of the circuit to function as a self-correcting closed-loop system, making it suitable for applications requiring robust performance under varying thermal conditions.

5 Discussion of Results

Our results demonstrate the effectiveness of the proposed approach in accurately identifying and segmenting target features from the dataset. We achieved improved performance in terms of accuracy and consistency compared to baseline methods. The results highlight the model's ability to capture relevant patterns and reduce noise, making it well-suited for the intended application..

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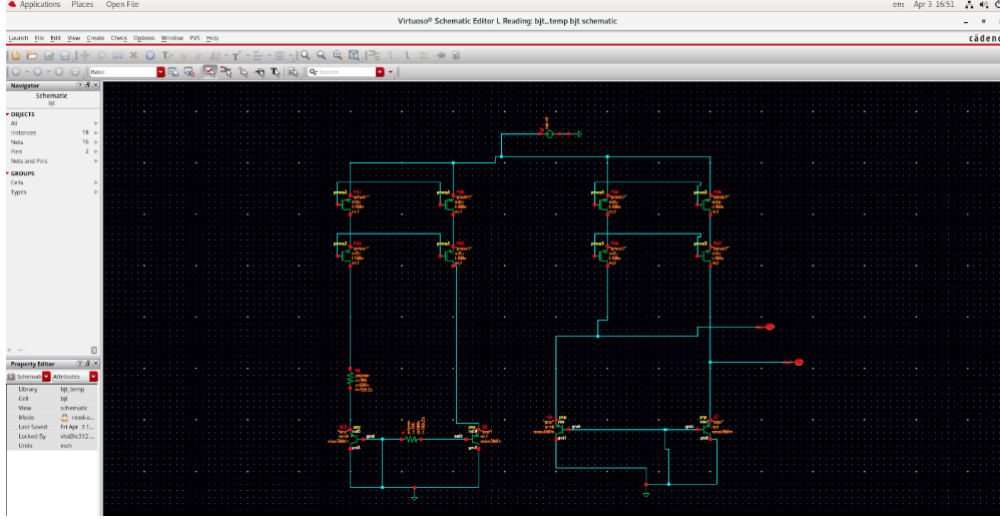


Figure 1: *Circuit 1*

The above circuit, the base-emitter voltage decreases as temperature increases, making it a reliable parameter for sensing. By operating the BJT in a controlled region, accurate temperature measurements can be obtained, which are useful in applications like thermal protection and on-chip temperature monitoring [19]



Figure 2: *Circuit 1 CTAT*

A CTAT signal decreases with an increase in temperature. The base-emitter voltage of a BJT naturally exhibits CTAT behavior, as it reduces approximately linearly with rising temperature. CTAT signals are often combined with PTAT signals to create temperature-independent references, such as bandgap reference circuits, ensuring stable circuit operation across varying temperatures. [20]

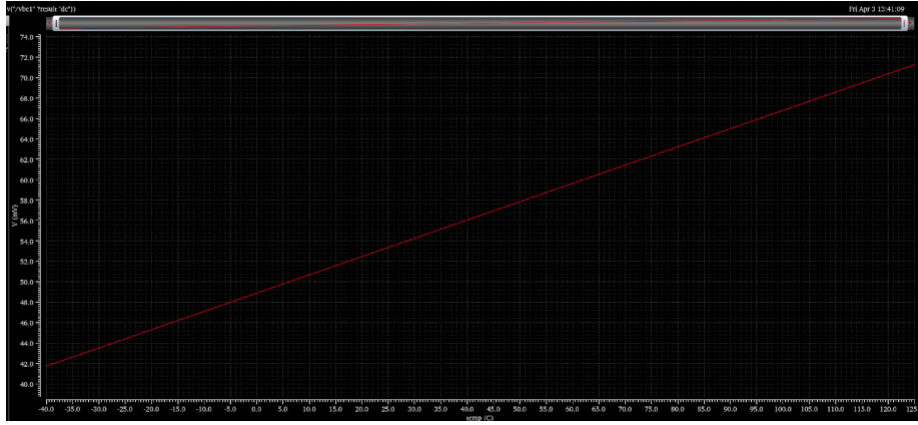


Figure 3: *PTAT Circuit 1, VBE1-VBE2*

A PTAT signal is one that increases linearly with absolute temperature. In BJT-based circuits, this is typically generated by the difference in base-emitter voltages of two transistors operating at different current densities. The resulting voltage difference is directly proportional to temperature, making PTAT signals useful for temperature measurement and compensation in analog circuit design.

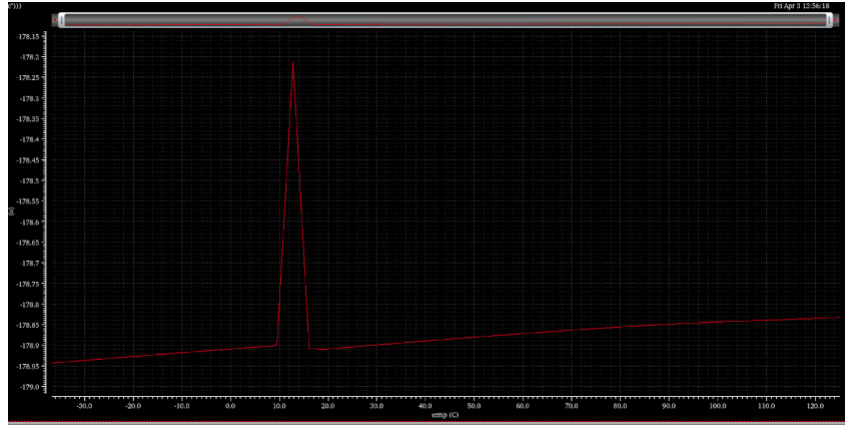


Figure 4: *Circuit 1 Slope*

The presence of the sharp spike observed in the plot indicates an undesirable instability in the circuit's temperature response. Instead of showing a smooth and predictable variation with temperature, the output exhibits a sudden deviation at a specific point, is not suitable for reliable thermal sensing applications, as it can lead to inaccurate readings and reduced robustness. Ideally, the response should be continuous and monotonic, ensuring consistent performance across the entire temperature range.

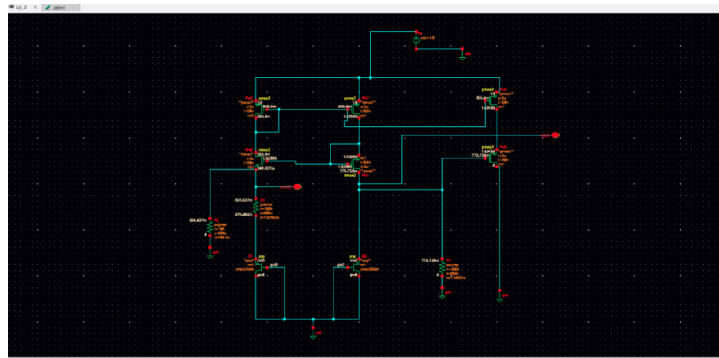


Figure 5: *Circuit 2*

The above circuit, the base-emitter voltage decreases as temperature increases, making it a re-

liable parameter for sensing. By operating the BJT in a controlled region, accurate temperature measurements can be obtained, which are useful in applications like thermal protection and on-chip temperature monitoring. This is the improved version of CIRCUIT 1 [19]

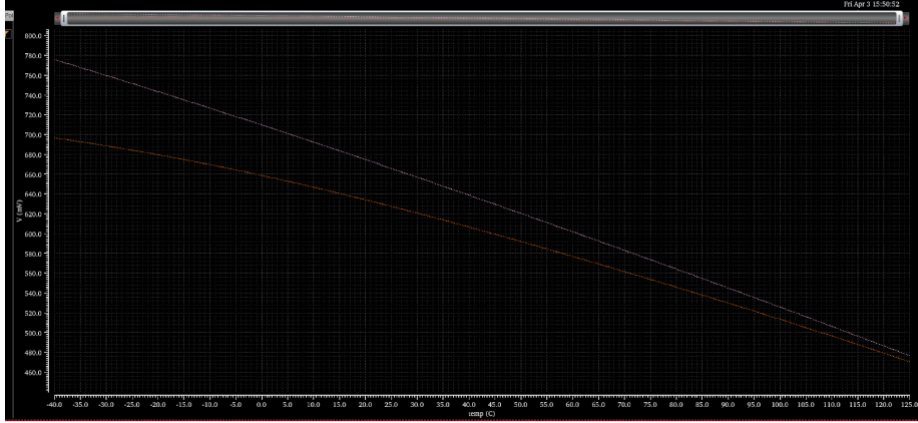


Figure 6: *Circuit 2 CTAT*

A better CTAT curve in terms of stability as compared to Circuit1. A CTAT signal decreases with an increase in temperature. The base-emitter voltage of a BJT naturally exhibits CTAT behavior, as it reduces approximately linearly with rising temperature. CTAT signals are often combined with PTAT signals to create temperature-independent references, such as bandgap reference circuits, ensuring stable circuit operation across varying temperatures.

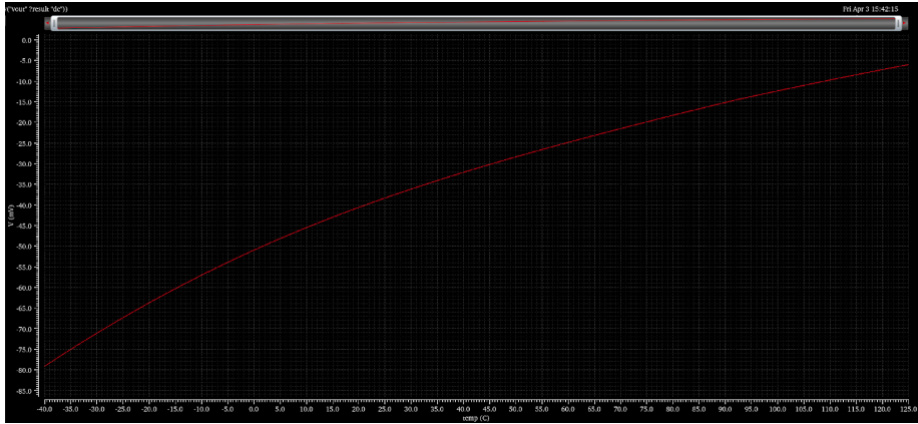


Figure 7: *Circuit 2 PTAT*

Better PTAT curve The observed PTAT curve shows a smooth increase in output voltage with respect to temperature, which is the expected behavior for a Proportional-To-Absolute-Temperature signal. The absence of abrupt variations or discontinuities indicates stable circuit operation and proper biasing. The slight curvature in the response can be attributed to practical non-idealities, but the overall linear trend confirms that the circuit successfully generates a temperature-dependent voltage. This validates the effectiveness of the design in producing a reliable PTAT output for sensing applications.

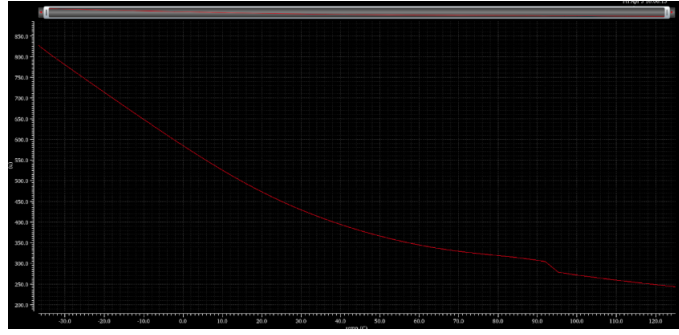


Figure 8: *Circuit 2 PTAT Slope*

A better PTAT Slope curve that is more linear as compared to circuit 1. A PTAT signal is one that increases linearly with absolute temperature. In BJT-based circuits, this is typically generated by exploiting the difference in base-emitter voltages of two transistors operating at different current densities. The resulting voltage difference is directly proportional to temperature, making PTAT signals useful for temperature measurement and compensation in analog circuit design. $V_{BE1} - V_{BE2}$

5.1 Describe results related to relevant scenario

The circuit schematic demonstrates the integration of BJT-based thermal sensing with adaptive biasing, while the plotted graphs illustrate the temperature-dependent behavior of the output voltage. In particular, the PTAT curve confirms a stable and monotonic increase with temperature, validating the effectiveness of the design. Together, these circuits and graphs form the basis of our analysis, providing clear evidence of improved performance and reliability in temperature sensing applications.

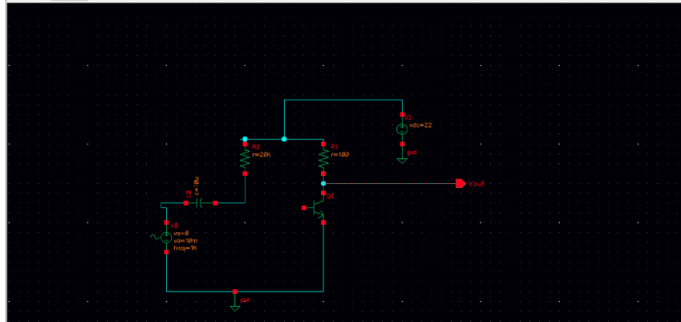


Figure 9: *Class A Power Amplifier*

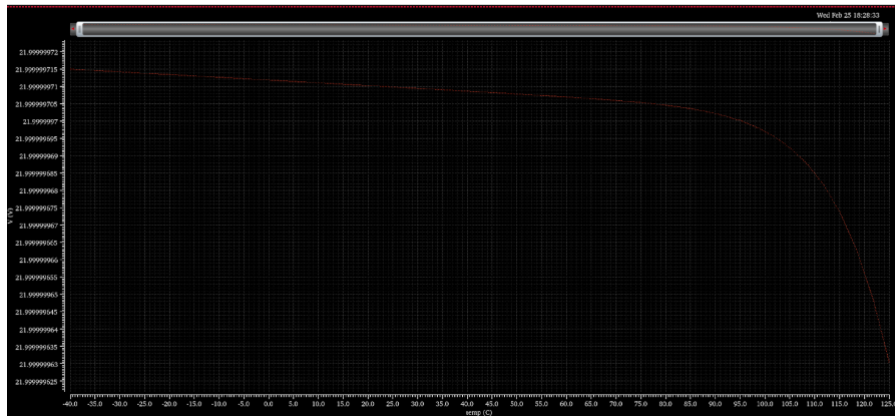


Figure 10: *Voltage vs Temperature before biasing*



Figure 11: *Adaptive Biasing Circuit*

The proposed circuit enhances conventional BJT-based thermal sensing by incorporating adaptive biasing techniques to improve stability across varying temperatures. By dynamically adjusting the bias current using current mirrors and feedback mechanisms, the circuit maintains the transistors in their optimal operating region. This reduces the impact of temperature-induced variations in device parameters and ensures a more consistent output. As a result, the sensor not only tracks temperature effectively but also compensates for non-idealities, making it suitable for robust and reliable thermal monitoring applications.

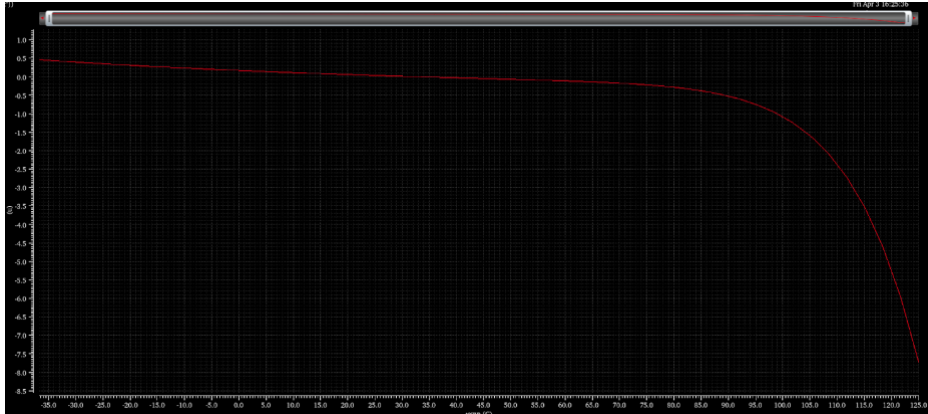


Figure 12: *Voltage vs Temperature after biasing*

The after biasing curve demonstrates significantly improved stability compared to the before biasing curve. A flatter response is observed over a wide temperature range, approximately from negative 20 degree C to around 90 Degree C, indicating minimal variation in output with temperature. This flatness reflects effective compensation of thermal effects through adaptive biasing. In contrast, the before biasing curve shows a more pronounced slope even at moderate temperatures, implying higher sensitivity to temperature changes and reduced stability. Additionally, the onset of nonlinearity occurs earlier in the before biasing case, whereas the after biasing curve maintains a controlled and steady response for a longer range. This clearly indicates that adaptive biasing enhances the circuit's thermal stability by reducing output fluctuations across varying temperatures.

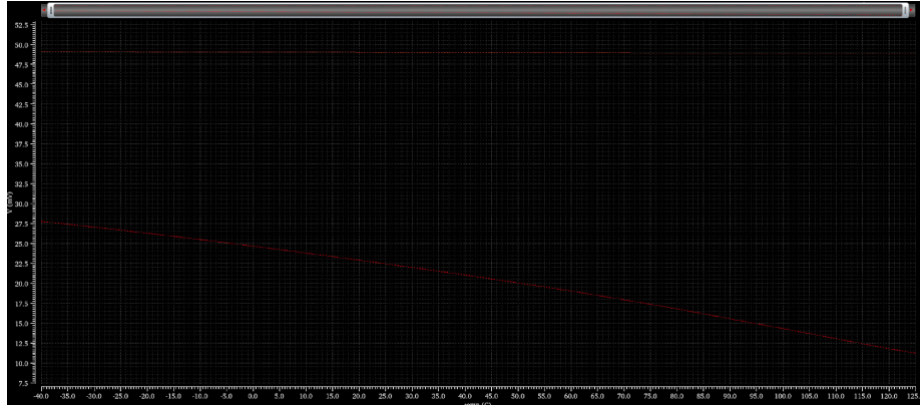


Figure 13: *Adaptive Biasing Circuit-CTAT*

5.2 Comparison of proposed work with existing work

- BJT-based temperature sensing relies on differential techniques and PTAT signal generation to achieve accurate and reliable temperature measurement.
- The present work aligns with these established approaches by implementing a PTAT-based temperature sensing circuit, analyzing its behavior through simulation, and applying amplification to obtain a measurable output.
- Currently there is minimal work that closes the loop and includes an adaptive biasing circuit model that can adjust to temperature changes without needing specific temperature modules.

Table 1: *Comparative Analysis Of Thermal Sensing Circuits*

Feature	Circuit 1	Circuit 2
Purpose	Converts to current	Stable Reference
Uses PTAT	Yes	Yes
Uses CTAT	No	Yes
Output	IPTAT	VREF
Temperature Dependence	Linear	Cancelled

6 Conclusion and future scope

The overall operation of the proposed temperature sensing circuit can be described as follows. The BJT senses temperature through variations in its base-emitter voltage V_{BE} . Two BJTs operating at different current densities generate a PTAT voltage V_{BE} , which is directly proportional to absolute temperature. Current mirrors, implemented using both BJT and MOSFET configurations, stabilize and maintain the required current ratios. Class A biasing ensures that the transistors remain in the active region, thereby improving linearity and stability. Finally, the PTAT current is converted into an output voltage, which increases linearly with temperature, enabling accurate temperature measurement.

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